Semantic Retrieval of Aviation Safety Reports with Frozen Sentence Embeddings

Project Milestone (Phase 2)

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***Abstract*—Safety analysts need to find past NASA ASRS reports that describe the same operational problem as a new one, even when pilots word it differently. We built one pipeline that indexes all 47,723 reports in a public ASRS release and scores keyword search (BM25), two frozen sentence encoders (all-MiniLM-L6-v2 and e5-small-v2), and rank-fusion hybrids on three evaluation tracks. The first track uses 1,000 analyst synopses as paraphrase queries. The second uses 40 analyst-style issue descriptions with graded relevance labels derived from NASA's coding ("silver" labels). The third is a 100-item test of whether meaning-changing edits (negation, numbers, direction, event order) fool the models. Contrary to our expectation, BM25 is the strongest single system (MRR@10 0.425 vs. 0.268 for MiniLM). Fusing BM25 with MiniLM gives the best topical ranking (nDCG@10 0.706 vs. 0.632, p=0.002). When word overlap is controlled, E5 tells a same-meaning sentence from a minimally edited one 85% of the time. Yet every system prefers the edited sentence when it shares more words (at most 17% correct), and event order stays near chance.**

I. Introduction

The Aviation Safety Reporting System (ASRS) holds hundreds of thousands of voluntary, free-text incident reports. Keyword search misses paraphrases ("climbed through our assigned altitude" vs. "overshot the level-off"), while frequent words such as "altitude" return thousands of off-topic hits. Sentence-BERT-style encoders [1] encode each passage once and compare by cosine, so they are fast enough to search a whole archive. They are trained on general English, however, and can blur details such as negation or event order that decide the meaning of a safety report [2]. Our research question is: *can a frozen sentence embedding retrieve ASRS reports that describe the same issue when the wording differs, and does the ranking fail under negation, altitude, direction or event-order changes?* By this milestone the data pipeline, all four proposed systems, and an evaluation harness are implemented, and we have first results on all three evaluation tracks.

II. Approach

A. Retrieval systems

BM25 [3] (k_1=1.2, b=0.75, Lucene IDF) on lowercased, stopword-filtered, Snowball-stemmed tokens. We implemented it on sparse matrices so that the same corpus statistics can also score arbitrary sentence pairs for the meaning-change test. A unit test checks that its scores equal those of the `bm25s` library. **Dense encoders** are used frozen through sentence-transformers. all-MiniLM-L6-v2 [4], [5] (22M parameters, 384-d, trained on 1B sentence pairs) and e5-small-v2 [6] (33M parameters, 384-d, `query:`/`passage:` prefixes) produce L2-normalised vectors, and search is exact (every chunk is scored). Narratives are longer than the encoders' context windows, so we encode *chunks* and score a report d by its best chunk (MaxP [7]):

$$s(q,d) = \max_{c \in d} \cos\big(f(q), f(c)\big), \tag{1}$$

returning $\arg\max_c$ as the matched passage shown to the analyst. **Hybrids** fuse two report rankings with reciprocal rank fusion [8], $s(d) = \sum_r 1/(60 + \mathrm{rank}_r(d))$, over the top 1,000 reports of each component.

B. What is ours

We wrote the data pipeline, the BM25 implementation, MaxP scoring and fusion, a config format in which adding a model takes one YAML entry, the query set and its labeling rules, the meaning-change test set, and all evaluation code (about 1,400 lines of Python, with 20 unit tests). We reuse sentence-transformers, Hugging Face `datasets`, scikit-learn and PyStemmer, and the two pretrained checkpoints.

III. Experiments

A. Data and pre-processing

We use `elihoole/asrs-aviation-reports` [9]: 47,723 de-identified reports (Jan. 2012–Mar. 2022) split into 38,655 train, 4,295 validation and 4,773 test. Each record has one or two reporter narratives (319 words on average, 15.2M words in total), an analyst synopsis (27 words), and NASA analyst codes. We search the narratives only. Pre-processing went through two versions; the steps of the current one (v2) are:

1) **Comma repair.** The export replaced every comma with a semicolon ("10;000 FT", "IAF; we set"). We restore commas in the free text, which is essential for the numeric test category. The coded fields keep ";" as their list separator.
2) **Placeholder removal.** 2,906 second narratives (26%) are editorial notes, e.g. "[Report narrative contained no additional information.]". We drop them (see §III-E).
3) **Incident grouping.** Union-find over shared accession numbers and identical narratives finds 70 reports that describe the same incident as another record; 13 such groups cross the official splits. Retrieving a sibling report counts as correct, and the planned fine-tuning will split by group.

4) **Chunking.** Narratives contain no paragraph breaks, so we pack sentences into windows of at most 150 words with a one-sentence overlap. A leftover tail under 40 words is merged into the previous window. This gives 136,842 chunks (2.9 per report; 150 WordPiece tokens on average, 99th percentile 222), which fit MiniLM's 256-token limit.

B. Evaluation

Synopsis track (automatic). The query is the analyst's synopsis of one of 1,000 validation-split reports (seed 13). Synopses are written separately from the narrative, so they are natural paraphrases. The relevant set is the summarised report plus its incident siblings. Metrics: MRR@10 and Recall@k. **Topical track.** We wrote 40 analyst-style queries (20 dev, 20 test) over 15 event types, 19 of them in the proposal's focus area (altitude, clearance, communication, heading). Until the human labels are done, each query has a rule over NASA's codes that assigns *silver* grades. A report gets grade 2 (same issue) when the event type and the specific mechanism both match, e.g. anomaly *Altitude Overshoot* and human factor *Distraction*. It gets grade 1 (related) when only the event family matches. Metrics are P@10 (grade 2), lenient P@10 (grade $\geq$1) and nDCG@10 with linear gains [10]. **Meaning-change test.** We wrote 100 items, 25 per category, each made of an original, a same-meaning paraphrase, and a minimally edited version whose meaning changes. For example, the negation item "The crew did not descend" has the paraphrase "The crew maintained altitude" and the edit "The crew descended". The proposal's condition anchors on the *original*. Since the edit shares almost all of the original's words, we added a controlled condition that anchors on the *paraphrase*, which overlaps the original and the edit about equally. An item passes when $\mathrm{sim}(\text{anchor}, \text{same meaning}) > \mathrm{sim}(\text{anchor}, \text{edited})$; chance is 50%. All retrieval means carry 95% bootstrap intervals, and systems are compared with paired randomization tests [11].

C. Experimental details

No model is trained. We ran everything on an Apple M1 Max (PyTorch MPS backend) with batch size 128. Indexing all chunks takes 6 s for BM25, 3.2 min for MiniLM and 5.3 min for E5. Answering 1,000 queries takes under 3 s per single system and 8 s per hybrid. Configs, per-query scores and run files are in the repository, so every number below can be regenerated with `asrsbench run configs/milestone.yaml`.

D. Results

Keyword search is the strongest single system (Table I). We expected the opposite. On synopsis queries BM25 ranks the right report first 36% of the time, against 21% for MiniLM and 19% for E5 (all pairwise MRR differences $p < 0.02$). We checked whether exact identifiers explain this. Only 34% of synopses repeat an identifier from their narrative (an airport code, aircraft type or runway). These help every system, but BM25 still leads on the other 66% (MRR@10 0.299 vs. 0.181). This agrees with the BEIR finding that BM25 is a strong zero-shot baseline outside the encoders' training domain [12]. A likely factor is that ASRS text is full of abbreviations (FL240, LNAV/VNAV, RA) that general-English encoders have rarely seen.

TABLE I
RETRIEVAL RESULTS (95% CI HALF-WIDTH $\approx$0.03 FOR SYNOPSIS, $\approx$0.08 FOR TOPICAL). TOPICAL LABELS ARE SILVER (FROM ANALYST CODES).

	Synopsis (n=1000)		Topical (n=40)		
System	MRR@10	R@100	nDCG@10	P@10	P@10$_{\geq 1}$
BM25	**.425**	.753	.632	.537	.705
MiniLM	.268	.648	.593	.468	.740
E5-small	.241	.607	.438	.352	.505
BM25+E5 (RRF)	.378	.759	.633	.537	.707
BM25+MiniLM (RRF)	.380	**.786**	**.706**	**.585**	**.798**

The two signals are complementary. BM25+MiniLM fusion improves topical nDCG@10 by 0.074 over BM25 (p=0.002) and by 0.113 over MiniLM ($p < 0.001$), and it has the best Recall@100 on synopses. MiniLM beats BM25 on lenient P@10 (0.740 vs. 0.705) but not on strict P@10 (0.468 vs. 0.537). It finds reports of the right event family but often not the right mechanism. The best (BM25+MiniLM) and worst (E5) systems are the same on the dev and test query halves. Altitude queries, the proposal's focus area, are the hardest for the encoders (nDCG@10: BM25 0.535, MiniLM 0.444, E5 0.159).

E5 is pulled toward short texts. E5 trails MiniLM on retrieval despite being the newer retrieval model. Its similarity scores for short units are *lower* on average but much more spread out (mean per-query z-score -0.83 for units under 30 words). Because MaxP takes a maximum over many chunks, that upper tail wins: short units supply 9.9% of E5's top-50 results but only 0.9% of the index, and MiniLM shows almost no such effect.

Meaning-change test (Fig. 1). In the proposal's condition, all systems prefer the one-word edit over the true paraphrase (accuracy 5–17%). Surface overlap dominates similarity for both keyword and embedding methods, so on this evidence a frozen encoder would rank "descended" above "maintained altitude" as a match for "did not descend". With overlap controlled, the encoders do encode much of the meaning: E5 0.85 and MiniLM 0.70 overall, with numbers at 0.96 for both. Event order remains the weak spot (MiniLM 0.40, E5 0.60), which fits mean pooling being largely order-insensitive. BM25 scores 0.22 on negation, partly because the standard stopword list deletes "not".

E. A pre-processing finding

In our first run (v1, without placeholder removal or tail merging), 18.5% of E5's top-10 matched passages had fewer than 30 words, against 3.7% of all chunks. Many were placeholders such as "[Narrative #2 contains no additional information.]". Pre-processing v2 cut short units to 0.9% of the

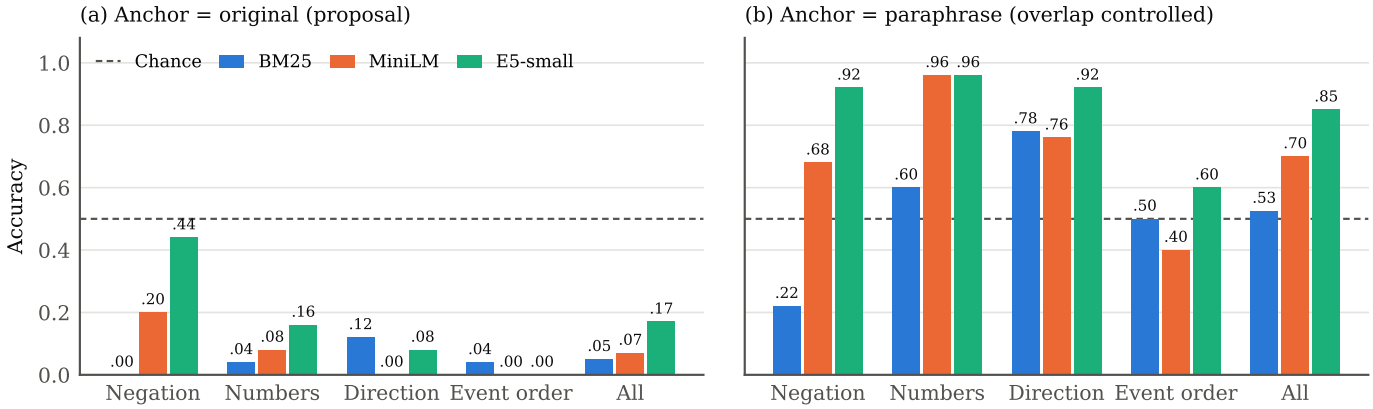


Fig. 1. Meaning-change test (25 items per category; 95% Wilson interval $\approx \pm 0.19$). (a) The edited sentence shares nearly all words with the anchor, and every system is fooled. (b) With word overlap balanced, the encoders mostly detect negation, number and direction changes, but not event order.

index, yet E5's topical score barely moved (0.444 to 0.438). The bias therefore comes from genuinely short reports, not just from the artifacts, which motivates the calibration work below.

IV. Future work

1) **Human labels (next two weeks).** `asrsbench pool` already merges the top-10 of every system into blind, shuffled judging sheets. Both of us will judge with a 25% overlap and report Cohen's κ, and we will extend the set toward the proposed 50–80 queries. Silver labels come from codes and synopsis keywords, so they may favour lexical matches; the human labels will test that.
2) **Length bias and report scoring.** We are running a whole-report versus best-chunk scoring ablation. Next we will test score calibration, which normalises each unit's score against its typical similarity to queries.
3) **More encoders.** The configuration already includes bge-small, all-mpnet-base and e5-base, and a BM25 variant that keeps negation words; those runs are in progress.
4) **Stretch goals**, in the proposal's order: report versions (B), a small fine-tune on near-miss pairs split by incident group (D), and ADS-B tracks (C).

The held-out topical test queries and the synopsis test split will be scored once, for the final report (Nov. 30). The talk on Nov. 23 will present the four-system comparison and failure examples.

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